PENGZHEN LI

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EDUCATION

University of Tennessee, Knoxville expected 2027
Ph.D. in Economics

Shandong University 2018
M.A. in Finance

Tongji University, Shanghai 2015
B.A. in Economics

RESEARCH INTERESTS

International Trade, Trade Policy, Applied Econometrics, and Political Economy.

WORKING PAPERS

Trade Restrictiveness Indices with Discriminatory Tariffs with James Lake [**Job Market Paper**]. *Presented at the Midwest Economic Theory and International Trade Meetings, Virginia Tech, 2025.*
Abstract: Trade-restrictiveness indices summarize a country's tariff schedule as a single welfare-equivalent uniform tariff, but the existing indices assume most-favored-nation tariffs. Since 2018 the largest tariff changes—the US–China tariff war, retaliatory tariffs, and proliferating preferential agreements—have been discriminatory by design. We extend the trade-restrictiveness index to partner-specific tariffs while preserving its welfare interpretation and derive a closed-form decomposition into a "level" component (how much a country restricts trade) and a "discrimination" component (how unevenly it does so). Computing the indices from WITS/Comtrade data for 2017–2024, we find that the MFN-only index systematically understates trade distortions when discrimination is high, and the decomposition separates, for the first time in this framework, the welfare cost of discrimination from the cost of protection.

A Mean-Field Game of Mayoral Endorsements: Identification and Evidence from Brazilian Coalitional Presidentialism with Matt Van Essen and Luiz Lima. *2024 J. Fred and Wilma Holly Award (best second-year paper).* [paper] [slides]
Abstract: A mayor's return from endorsing a presidential candidate falls as other mayors endorse the same one, so endorsement is a strategic discrete-choice game that has gone structurally unestimated: among Brazil's 5,570 mayors the joint action space is too large to enumerate. Anonymity collapses the problem. When payoffs depend on rivals only through aggregate shares, equilibrium is a fixed point on the candidate simplex whose dimension does not grow with the number of players, and the model estimates by least squares in milliseconds. Estimating it on Brazil's 2014 election, we then show what the exercise does not buy: the prize parameter is algebraically the average log-odds of alignment rescaled by the share of mayors classified as aligned, so it changes sign with the classification rule. What survives is reduced form. A close-race difference-in-differences design puts discretionary transfers to co-partisan mayors 10.6 percent above those of other coalition members and finds no premium on the formula-based transfer that leaves no discretion to allocate.

Price Risk, Irreversible Investment, and the Welfare Benchmark for the Section 45Z Clean Fuel Production Credit with T. Edward Yu. [paper] [slides]

How Should 45Z Pay for Risk? A Welfare-CVaR and Project-Finance Decomposition of the IRA Clean Fuel Production Credit with T. Edward Yu. [paper] [slides]

Estimating Pork Primal Price Relationships Using a Threshold Vector Autoregressive Analysis with Charles Martinez, Christopher N. Boyer, T. Edward Yu, and David Anderson. *SSRN working paper 4121953 (2022).* [paper]

WORK IN PROGRESS

A Weight-Free Benchmark for Multilateral Tariff Bargaining with Georg Schaur. *Draft, August 2026.* [paper] [slides]

PEER-REVIEWED PUBLICATIONS

"Spatial optimization of the sustainable aviation fuel supply chains from forest residues via fast pyrolysis/hydrotreatment considering feedstock ash content variability." **P. Li**, T. E. Yu, N. Labbé, N. Abdoulmoumine, M. Garcia-Perez, K. P. Hoyt, and B. C. English. *Biomass and Bioenergy*, 208 (2026), 108793.

"Assessing the impact of preprocessing and conversion technologies on the sustainable aviation fuel supply from forest residues in the Southeast USA." **P. Li**, T. E. Yu, C. Trejo-Pech, J. A. Larson, B. C. English, and D. N. Lanning. *Transportation Research Record*, 2678(9) (2024), 639–654.

"Beef price spread relationship with processing capacity utilization." C. Martinez, **P. Li**, C. N. Boyer, T. E. Yu, and J. G. Maples. *Journal of the Agricultural and Applied Economics Association*, 2(1) (2023), 133–145.

"Do green technology innovations contribute to carbon dioxide emission reduction? Empirical evidence from patent data." K. Du, **P. Li**, and Z. Yan. *Technological Forecasting and Social Change*, 146 (2019), 297–303.

PRESENTATIONS

Midwest Economic Theory and International Trade Meetings, Virginia Tech — Spring 2025
Western Regional Science Association, Annual Meeting, Hawaii — 2023
Western Agricultural Economics Association, Annual Meeting, Santa Fe, NM — 2022
Southern Agricultural Economics Association, Annual Meeting, New Orleans, LA — 2022

TEACHING EXPERIENCE

University of Tennessee

Instructor

ECON 351: Monetary Economics — Spring 2025, Spring 2026
ECON 322: The Global Economy: Trade and Development — Summer 2025
ECON 313: Intermediate Macroeconomics — Summer 2026

Teaching Assistant

Ph.D. courses:
ECON 511: Microeconomic Theory — Fall 2023, Fall 2024, Fall 2025
ECON 512: Advanced Microeconomics — Spring 2024
ECON 581: Mathematical Methods in Economics — Summer 2024
Master's courses:
AREC 525: Agribusiness Operations Research Methods — Spring 2022, Spring 2023

PROFESSIONAL EXPERIENCE

University of Tennessee – Knoxville, TN 2020 – present

Graduate Research & Teaching Assistant. USDA NIFA and FAA ASCENT projects (2020–2023): mixed-integer supply-chain optimization (GAMS), IMPLAN/DEA siting analysis, and spatial data pipelines (ArcGIS, R, Python) for forest-residue sustainable aviation fuel; two publications and three working papers.

Shandong University – Jinan, China 2016 – 2018

Graduate Research Assistant. Panel-threshold and machine-learning analysis of green innovation and CO_2 emissions (published in *TFSC*, 2019).

HONORS AND AWARDS

Charles B. Garrison Award for Excellence in Teaching, University of Tennessee	2026
Graduate Summer Fellowship, University of Tennessee	2025
J. Fred and Wilma Holly Award (best second-year paper), University of Tennessee	2024
J. Fred and Wilma Holly Fellowship, University of Tennessee	2023 – present
Graduate Assistantship, University of Tennessee	2020 – present
Outstanding Volunteer, migrant-family tutoring program, Shanghai	pre-Ph.D.

SKILLS

Programming Languages: Stata, R, Python, SQL, GAMS, MATLAB, LaTeX, ArcGIS, IMPLAN.
Methods: panel and causal econometrics (FE, IV, DiD, event study, SCM, GMM, panel threshold), time series (TVAR, VECM, mGARCH), mixed-integer and stochastic programming, DEA, cooperative and mean-field games, machine learning.
Credentials: CFA Level II.
Languages: Chinese (native), English (fluent).

REFERENCES

James Lake
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Department of Economics
University of Tennessee
Email: jlake7@utk.edu

Georg Schaur
Leigh Burch III Economics Professor and Haslam Family Faculty Fellow
Department of Economics
University of Tennessee
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Matthew Van Essen
Lena M. and Clarence A. Garvin, Sr. Professor and Dr. Joe Johnson Faculty Fellow
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